INCH POUND
MIL-PRF-3098/17H
16 April 2015
SUPERSEDING
MIL-PRF-3098/17G
5 March 2010

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR37/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 90 kHz to 250 kHz; fundamental; noncontrolled; anti-resonance.

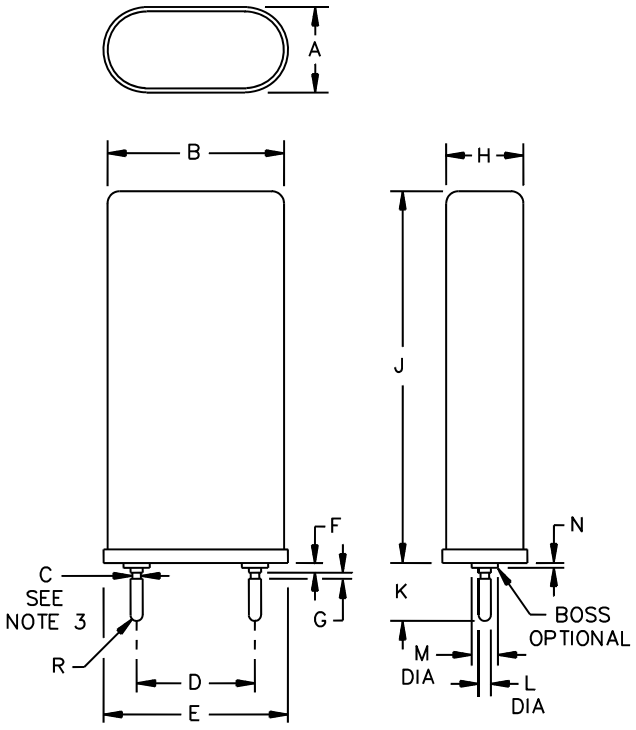

FIGURE 1. Crystal unit - CR37/U.

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.352	---	8.94
B	---	.725	---	18.41
C	.030	.037	0.76	0.94
D	.478	.494	12.14	12.55
E	---	.757	---	19.23
F	.030	.040	0.76	1.02
G	.015	.025	0.38	0.63
K	.223	.248	5.66	6.30
H	---	.317	---	8.05
J	---	1.526	---	38.76
L	.048	.052	1.22	1.32
M	.075	.141	1.90	3.58
N	.015	.025	0.38	0.63

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR37/U - Continued.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 90 kHz to 250 kHz, inclusive.

Capacitance, shunt: See table I.

Frequency tolerance:

Operating temperature range: ±200 parts per million (ppm).

Equivalent resistance: 90 kHz to 170 kHz: 5,000 ohms, maximum.
170+ kHz to 250 kHz: 5,500 ohms, maximum.

Mode of oscillation: Fundamental.

Antiresonance, load capacitance: 20.0 pF ±0.5 pF.

Operating temperature range (noncontrolled): -40°C to +70°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Shock (specified pulse):

Frequency change permitted: ±10 ppm.

Equivalent resistance change permitted: ±15 percent.

Vibration: Method 201 of MIL-STD-202, amplitude 0.015 inch (total excursion 0.030 inch).

Frequency change permitted: ±10 ppm.

Equivalent resistance change permitted: ±15 percent.

Temperature run:

Frequency change permitted: ±10 ppm.

Equivalent resistance change permitted: ±15 percent.

Bond strength: 90 kHz to 170 kHz: 800 grams, minimum
170+ kHz to 250 kHz: 700 grams, minimum

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TABLE I. Capacitance, shunt.

Frequency range,	Capacitance 1/
kHz	pF
90 to 170	$\frac{450}{f} + 1.2$
170+ to 250	$\frac{322}{f} + 1.2$

1/ The letter “f” represents specified frequency in kHz. A capacitance tolerance of ±15 percent is permitted.

Referenced documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202.

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air force - 99
DLA – CC

Review activities:
Army - AR, MI,
Navy - AS, MC, SH
Air Force - 19, 84

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-2015-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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